Riva Harsh Mohta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.19
Beacon High School Bhosari Pune DT MR	Class XII (CBSE)	2024	94.60%
Euro School, Pune	Class X (CISCE)	2022	98.50%

POSITIONS OF RESPONSIBILITY

Coordinator- PR, Marketing and Outreach Domain | Finance and Consulting Club, IIT Hyderabad

- Collaborated with a team of 9 coordinators to plan and execute public relations, marketing, and outreach activities for club events and initiatives.
- Coordinated with 11 associates to support communication campaigns, promotional activities, and participant engagement, contributing to the successful outreach of club programs.

Core-MedTech Domain | Elektronica(Electrical and Electronics Club), IIT Hyderabad

- Studied ECG and EMG signal characteristics and applied MATLAB for signal visualization and basic analysis, strengthening foundational skills in biomedical signal processing.

Core | Cepheid (Astronomy and Astrophysics Club), IIT Hyderabad

- Participated in telescope observation sessions and astronomy outreach activities conducted by the club for students and astronomy enthusiasts.
- Attended learning sessions and discussions on observational astronomy, and space science, expanding knowledge of fundamental astronomical concepts.

SKILLS

Technical: Python, MATLAB, C/C++, Arduino, Biomedical Signal Fundamentals, Data Structures, Algorithms

Management/Business: Public Relations, Outreach Coordination, Event Promotion, Marketing Support

Collaboration & Communication: Teamwork, Presentation Skills, Technical Writing,Interpersonal Communication

RELEVANT COURSES

Programming & Computing: Data Structures & Applications, Algorithms and Data Structure Lab, Introduction to Programming, Introduction to Artificial Intelligence

Certifications: MATLAB Onramp(*MathWorks*), Signal Processing Onramp(*MathWorks*), AI Basics: Learn Artificial Intelligence from Scratch(*Padikkable AI Systems-Udemy*)

Biomedical Engineering: Biomedical Imaging, Bioengineering, Mathematical Model and System Biology, Medical Nanotechnology-I, Basic Bioinformatics

Signal Processing & Mathematics: Linear Systems and Signal, Transform Techniques, Differential Equations, Elementary Linear Algebra, Introduction to Probability, Introduction to Statistics, Complex Variables

PROJECTS

IV Flow Monitoring System using Ultrasonic Sensor

- Developed a non-contact IV flow monitoring system using Arduino and an HC-SR04 ultrasonic sensor, with one other person, to track fluid-level variations and estimate flow conditions.
- Implemented averaging, threshold-based detection, and time-window analysis techniques to identify slow-flow and stopped-flow scenarios while reducing measurement noise.
- Integrated a buzzer alert mechanism for abnormal flow conditions and evaluated system performance under varying fluid levels to improve monitoring reliability.

Seating Posture Monitoring System using Ultrasonic Sensor

- Developed a real-time seating posture monitoring system using Arduino and an HC-SR04 ultrasonic sensor, with one other person, to measure user distance and classify sitting posture based on predefined threshold ranges.
- Implemented averaging techniques and threshold-based posture classification to improve measurement stability and accurately detect slouching and absence conditions.
- Integrated buzzer-based alerts for posture correction and evaluated system performance across different sitting positions to optimize detection reliability.

Gas Turbine Emission Prediction using ANN

- Built a 9-16-16-2 feed-forward ANN for multi-output CO/NOx regression using 7,384 real-world sensor samples.
- Performed architecture and hyperparameter optimization across hidden layers, neurons, ReLU/Sigmoid/Tanh activations, epochs and training-set size; the 32-16 ReLU architecture achieved the best tested performance (R^2 = 0.844, MSE = 0.00102).
- Compared Adam vs RMSProp, with Adam providing better test performance (R^2 0.840 vs 0.817) and more stable convergence.

EXTRACURRICULAR

Dronalaya Career Booster Programme, MVPM, Pune

10-day residential programme focused on leadership, communication, professional development, goal setting, and industry exposure.

Completed A2-level German (Goethe-Institut curriculum).